Extended Appendix

How Do We Hear a Rāga? An Interpretable Cognitive Model of Melodic Recognition

Krishnasree Kelath (Student ID: 2540124)

This appendix supplements the 10-page project report with mathematical definitions, per-stage diagnostics, per-rāga feature visualisations, all confusion matrices, and experimental results. All figures are generated by the pipeline scripts in the GitLab repository.

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1 Mathematical Definitions

This section defines every mathematical operation used in the pipeline.

1.1 Cents Conversion

The cent is a logarithmic unit of pitch interval where one octave equals 1200 cents. Given a fundamental frequency f_0 and tonic frequency Sa_{Hz}:

$$c = 1200 \times \log_2\left(\frac{f_0}{\text{Sa}_{\text{Hz}}}\right)$$

Example: If Sa = 150 Hz and $f_0 = 225$ Hz, then $c = 1200 \times \log_2(1.5) = 1200 \times 0.585 = 702$ cents (Pa, the perfect fifth). The logarithmic scale ensures that equal cent intervals sound perceptually equal regardless of register.

1.2 Octave Folding

All pitches are folded into a single octave $[0, 1200)$ using modular arithmetic:

$$c_{\text{mod}} = c \bmod 1200$$

This implements octave equivalence: Sa at 150 Hz and Sa at 300 Hz both map to 0 cents.

1.3 Circular Distance (Swara Mapping)

The 12 swara bins are arranged on a circle. The distance from a pitch c to bin b is:

$$d(c, b) = \min\big(|c - b|,\ 1200 - |c - b|\big)$$

This handles the octave boundary correctly: a pitch at 1190 cents is 10 cents from Sa (0/1200), not 1190 cents away. If $d(c, b) \leq 40$ cents, the frame is assigned swara b; otherwise it is labelled "glide."

1.4 Harmonic Tonic Scoring

For each tonic candidate frequency, a pitch-class histogram h is built (120 bins, 10-cent resolution). The score is:

$$\text{score} = \sum_{|i|<30} h[i] \ + \ 0.5 \times \sum_{|i-702|<30} h[i]$$

where the first term is Sa mass (0 cents) and the second is Pa mass (702 cents). The 0.5 weight reflects the tānpura drone structure: Sa is the primary tone, Pa is secondary reinforcement.

1.5 Weighted F1 Score

$$F_{1,\text{weighted}} = \sum_{i=1}^{K} \frac{n_i}{N} \times F_{1,i}$$

where n_i is the number of samples in class i, N is total samples, and $K = 13$ classes. Chosen over accuracy (which rewards ignoring minority classes) and macro F1 (unstable with small classes).

1.6 Permutation Test (H4)

The observed within/between-family confusion ratio is:

$$r_{\text{obs}} = \frac{\overline{C}_{\text{within}}}{\overline{C}_{\text{between}}} = 5.19$$

For $n = 10{,}000$ iterations, family labels are randomly shuffled and r_{perm} recomputed. The p-value is the fraction of permutations where $r_{\text{perm}} \geq r_{\text{obs}}$. Result: $p < 0.0001$ (0 of 10,000 permutations exceeded the observed ratio).

2 Pipeline Overview

2.1 Architecture

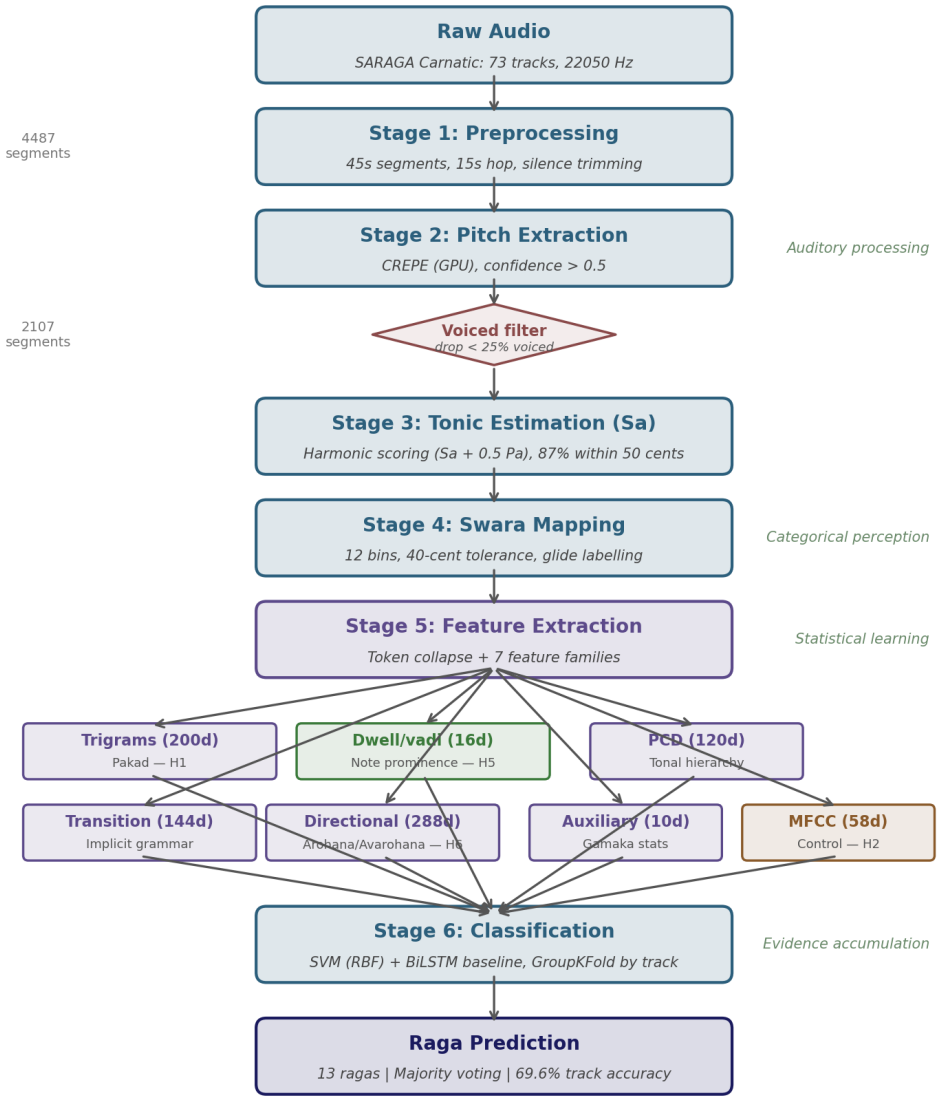


Figure 1: Six-stage pipeline architecture. Each stage mirrors a cognitive or musicological process. The seven feature families feed independently into the SVM classifier; Dwell/vādi (16d) achieves the best track-level weighted F1.

2.2 Dataset Distribution

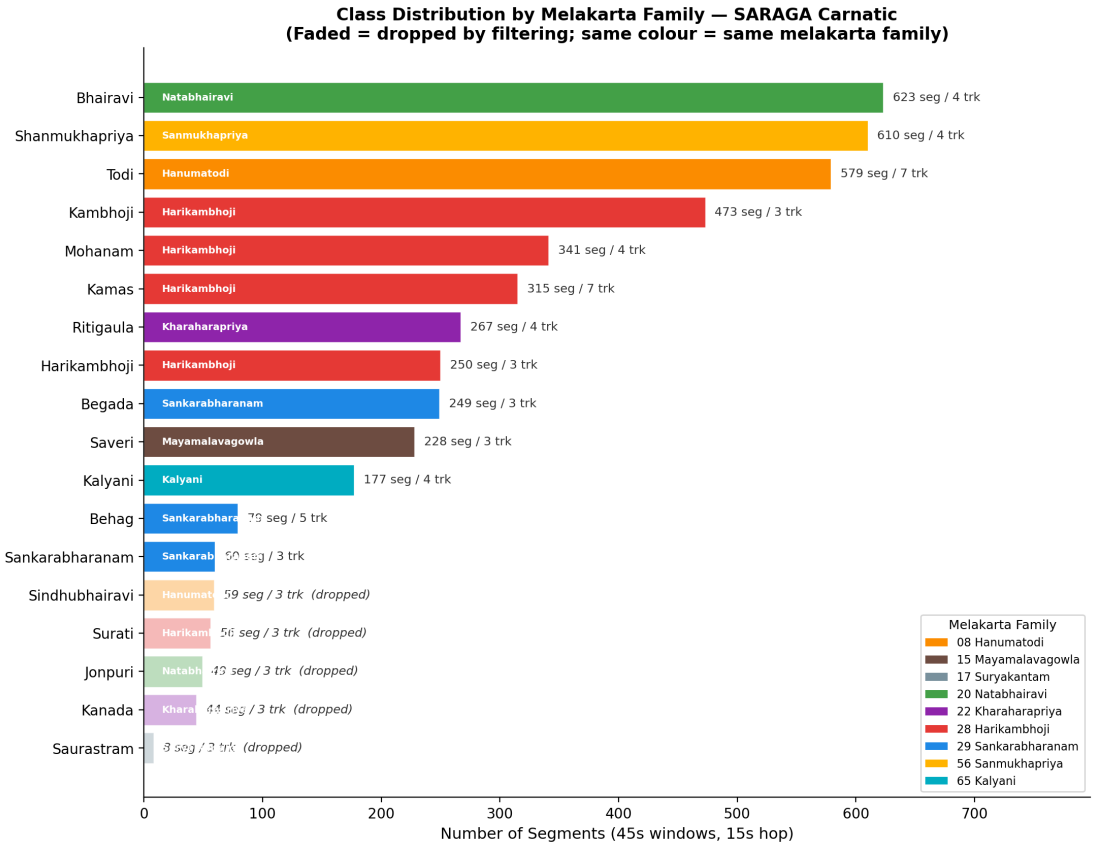
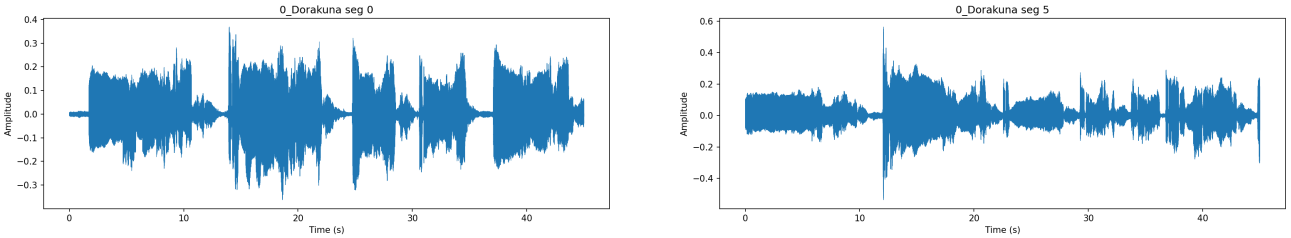


Figure 2: Segment counts for all 18 rāgas in the SARAGA Carnatic subset, coloured by melakarta (parent scale) family. Faded bars indicate rāgas dropped during filtering (< 20 segments after voiced-ratio and bad-tonic exclusion). The final evaluation set retains 13 rāgas across 46 tracks (1,648 segments) with a $\sim$10:1 imbalance between the largest (Bhairavi, 623) and smallest (Śankarābharaṇaṃ, 60) classes. The five-member Harikāmbhōji family (red) dominates both in membership and in within-family confusion (see §7.3).

3 Stage 2: Pitch Extraction

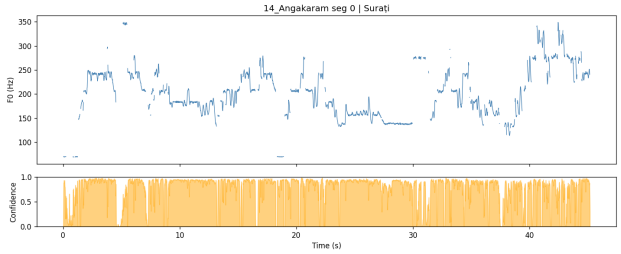
Representative CREPE pitch extraction outputs showing the extracted F0 contour (top) and voicing confidence (bottom) for two different tracks. These plots verify that the pitch tracker captures continuous gamaka contours and correctly masks unvoiced regions.



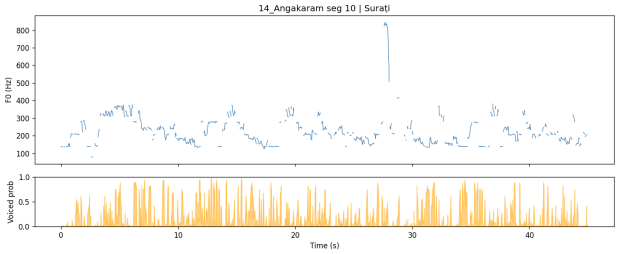
(a) Dorakuna, segment 0

(b) Dorakuna, segment 5

Figure 3: Pitch extraction diagnostics for Dorakuna. The F0 contour (top) shows continuous pitch movement characteristic of Carnatic vocal gamaka. The confidence trace (bottom) cleanly separates voiced melodic regions from silence/percussion.



(a) Angakaram, segment 0

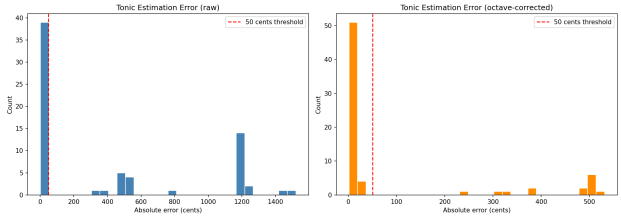


(b) Angakaram, segment 10

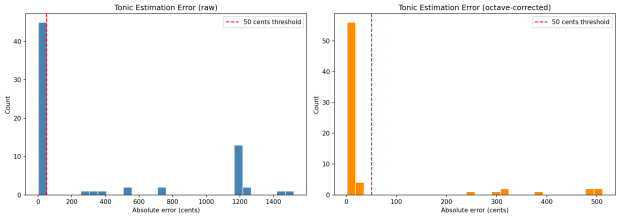
Figure 4: Pitch extraction diagnostics for Angakaram. Note the wider pitch range and more frequent voicing gaps compared to Dorakuna, reflecting differences in performance style and accompanying instrumentation.

4 Stage 3: Tonic Estimation

4.1 Baseline Results



(a) Sa-mass scoring (79.7% within 50 cents)



(b) Harmonic scoring (87.0% within 50 cents)

Figure 5: Tonic error histograms. Harmonic scoring produces a tighter distribution around zero cents error by exploiting the tānpura's Sa+Pa drone structure.

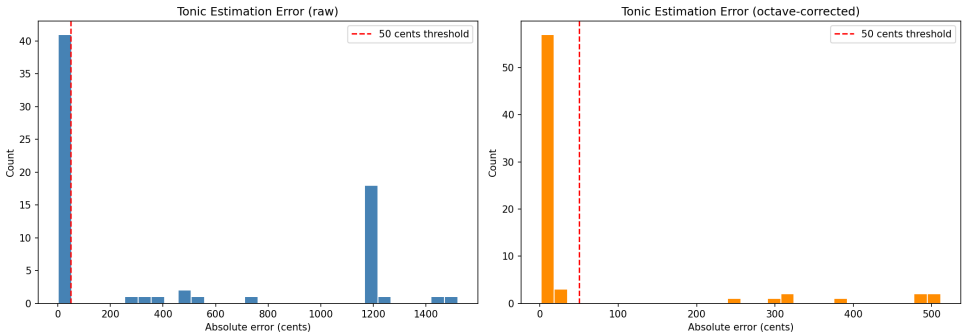


Figure 6: Smoothed harmonic scoring ($\sigma = 2$ bins): reduces narrow-peak sensitivity but does not improve overall accuracy. More octave errors than unsmoothed harmonic.

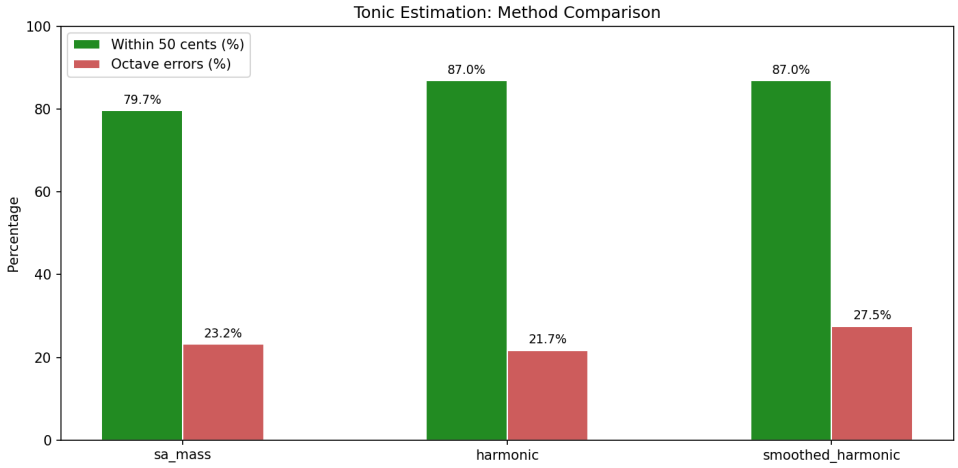
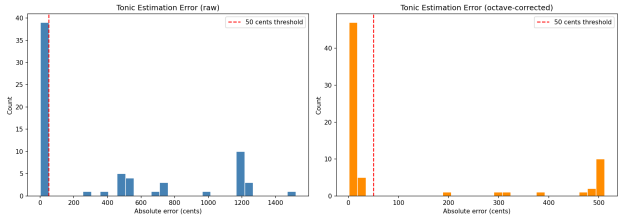


Figure 7: Head-to-head comparison of all three tonic scoring methods. Harmonic scoring achieves the highest within-50-cents accuracy (87.0%) with the fewest octave errors, confirming that the tānpura's Sa+Pa drone structure provides a reliable perceptual anchor for tonic estimation.

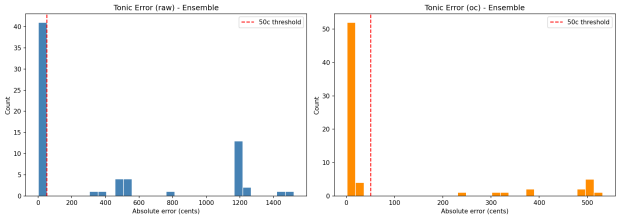
4.2 Tonic Experiments (Including Failed)

The following experiments attempted to improve tonic estimation beyond the harmonic baseline. Scripts are in `carnatic_scripts/experiments/`.

- **Harmonic triad scoring** (`03_tonic_baseline_with_penalty.py`): Extended Sa+Pa to Sa+Ma+Pa with a Pa>Sa penalty. Marginal improvement, confirming that the simpler Sa+Pa scoring already captures the dominant cue.
- **Ensemble voting** (`03_tonic_ensemble.py`, `v2`): Combined multiple scoring methods via weighted voting. Did not exceed the best single method.
- **Segment-level voting** (`03_tonic_segment_voting.py`): Per-segment tonic estimates aggregated via octave-invariant majority vote. Reduced outlier sensitivity but did not improve median accuracy.
- **Swara-validated tonic** (`03_tonic_swara_validated.py`): Used swara distribution quality as a secondary validation signal. Circular dependency (swara mapping requires a tonic) limited effectiveness.



(a) Triad scoring error distribution



(b) Ensemble voting error distribution

Figure 8: Tonic experiment error histograms. Neither the triad extension nor the ensemble approach substantially improved over the harmonic baseline (87.0% within 50 cents).

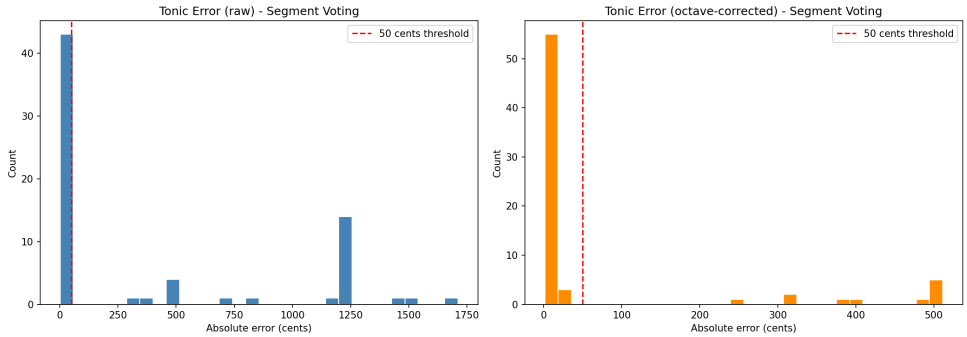
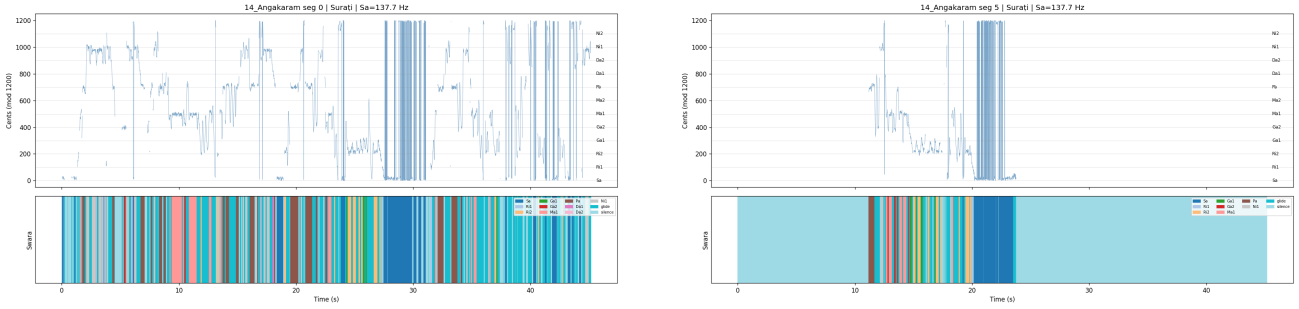


Figure 9: Segment-level voting tonic estimation. Per-segment estimates are aggregated via octave-invariant majority vote. Reduces outlier sensitivity but does not improve median accuracy over the track-level harmonic baseline.

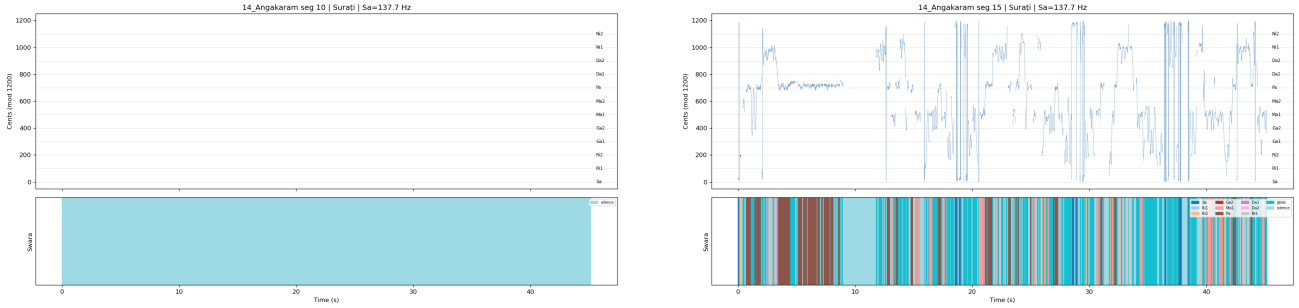
5 Stage 4: Swara Mapping

Swara mapping overlays the discrete swara labels (bottom, colour-coded) onto the continuous pitch contour in cents (top). These plots verify that the 40-cent tolerance correctly assigns swaras while labelling inter-note gamaka movement as "glide."



(a) Angakaram, segment 0 (b) Angakaram, segment 5

Figure 10: Swara mapping diagnostics. Top: pitch contour in cents (mod 1200) with horizontal lines at each swara bin. Bottom: colour-coded swara labels. "Glide" regions (visible between discrete swara assignments) correspond to ornamental gamaka movement, comprising ∼28% of voiced frames across the dataset.



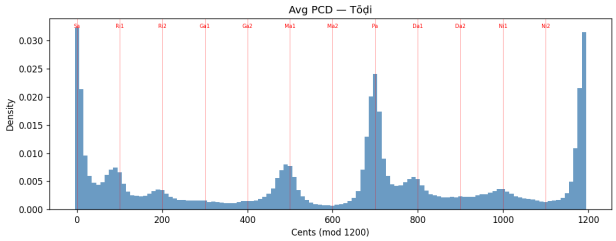
(a) Angakaram, segment 10 (b) Angakaram, segment 15

Figure 11: Additional swara mapping examples. The minimum-duration filter (100 ms) suppresses rapid label flickering from pitch estimation noise, visible as consolidated colour blocks rather than single-frame assignments.

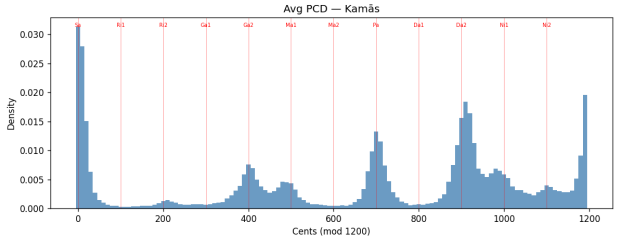
6 Stage 5: Feature Extraction

These figures show per-rāga pitch class distributions (120-bin histograms), swara transition matrices (12×12 heatmaps), and directional features. They demonstrate that rāgas with distinctive profiles classify well, while Harikāmbhōji-family rāgas with overlapping profiles are confused.

6.1 Pitch Class Distributions

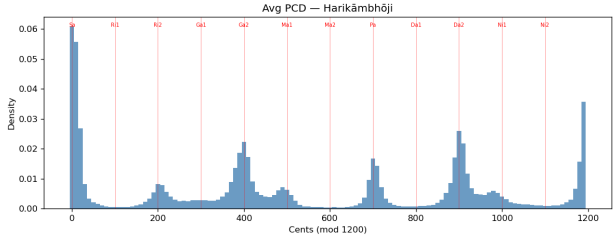


(a) Tōḍi (0.86 F1) — unique intervallic signature

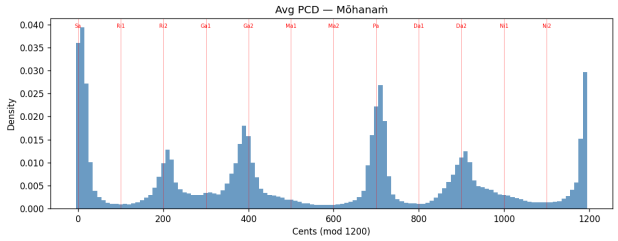


(b) Kamās (0.36 F1) — overlaps with Harikāmbhōji family

Figure 12: Pitch class distributions. Tōḍi's distinctive Ri1–Ga1 peaks explain its high classification performance. Kamās's PCD is nearly identical to its parent melakarta, explaining the family confusion.



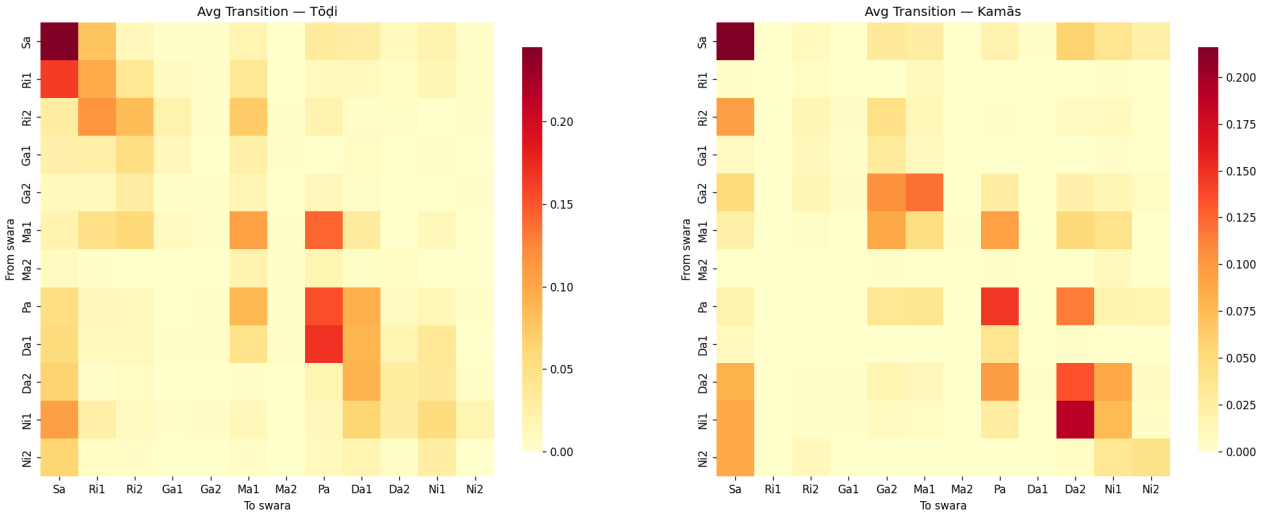
(a) Harikāmbhōji (parent, 0% F1)



(b) Mōhanaṁ (pentatonic, janya of Harikāmbhōji)

Figure 13: Harikāmbhōji (parent) vs Mōhanaṁ (child). The parent's PCD is a superset of the child's — every peak in Mōhanaṁ also appears in Harikāmbhōji, explaining the family confusion that drives H4. Mōhanaṁ's identity lies in what is *absent* (no Ma, no Ni), but our features encode presence, not absence.

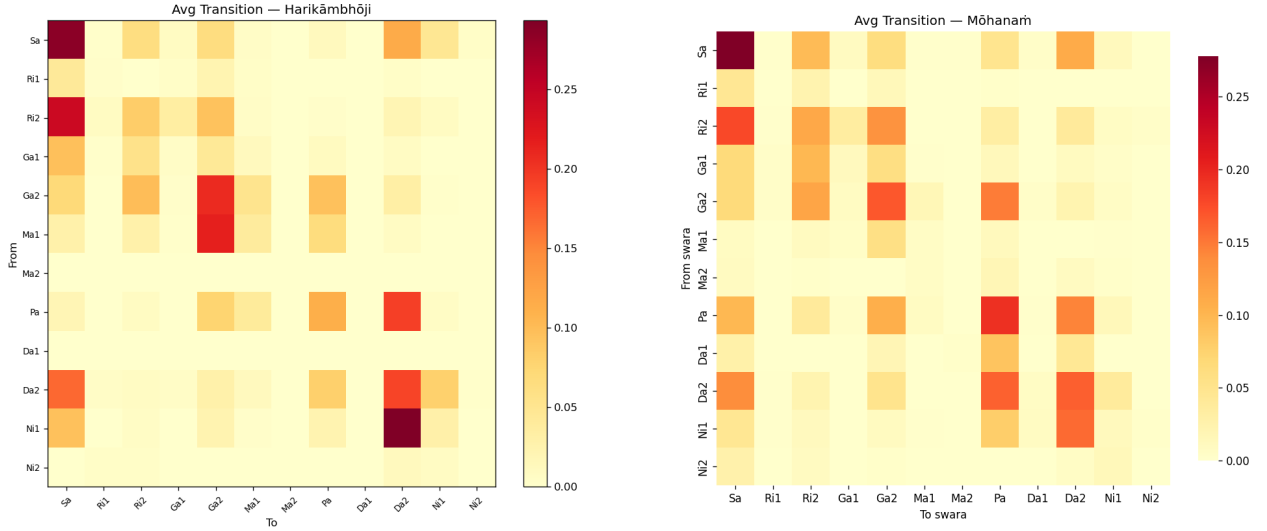
6.2 Transition Matrices



(a) Tōḍi — concentrated transitions

(b) Kamās — diffuse, shared with family

Figure 14: Swara transition matrices showing $P(\text{next} \mid \text{current})$. Tōḍi's transitions are concentrated along specific paths; Kamās shows distributed patterns typical of the Harikāmbhōji family.

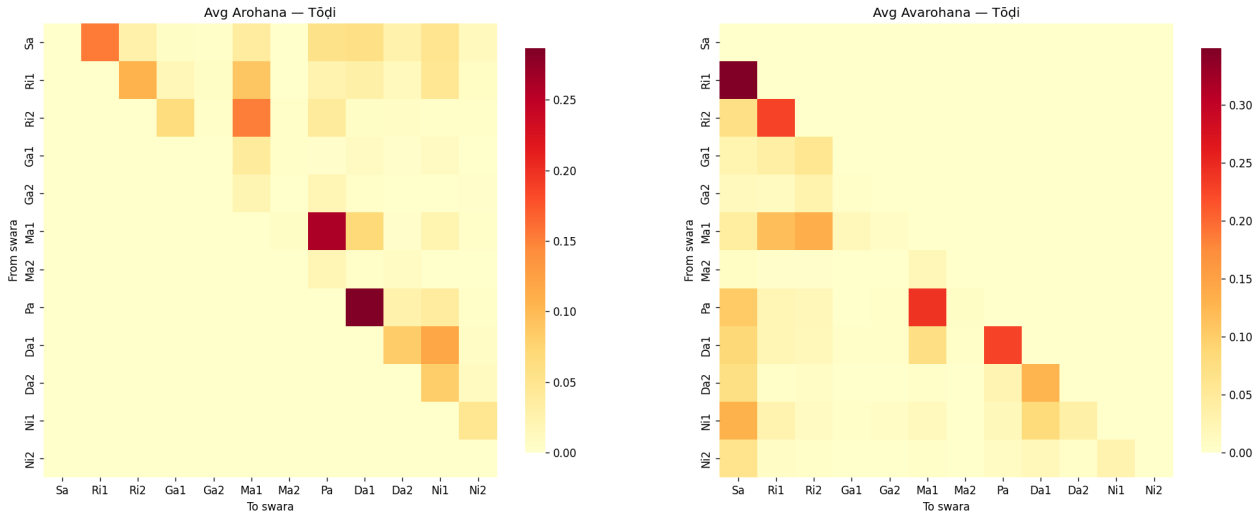


(a) Harikāmbhōji (parent, 0% F1)

(b) Mōhanaṁ (pentatonic child)

Figure 15: Harikāmbhōji vs Mōhanaṁ transitions. The parent's matrix is a dense superset; the child's is a sparse 5×5 subset. Kamās's transitions (above) are nearly identical to the parent's — the root cause of within-family confusion.

6.3 Directional Features (Ārohana / Avarohana)

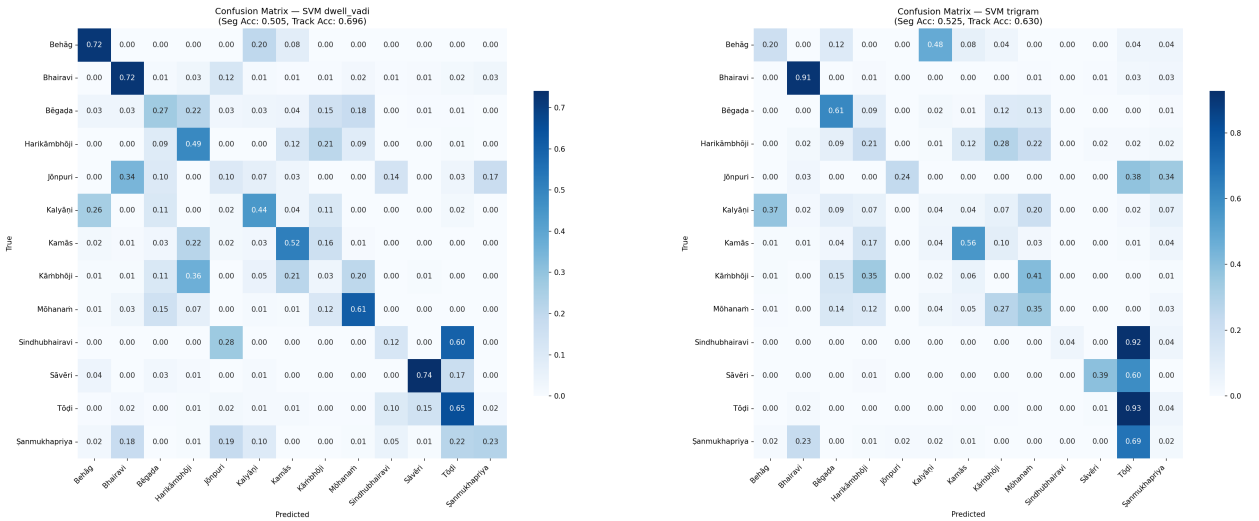


(a) Tōḍi ascending (ārohana) (b) Tōḍi descending (avarohana)

Figure 16: Directional PCD for Tōḍi. Tested under H6; directional encoding did not improve classification because gamakas blur the ascending/descending distinction in performed Carnatic music.

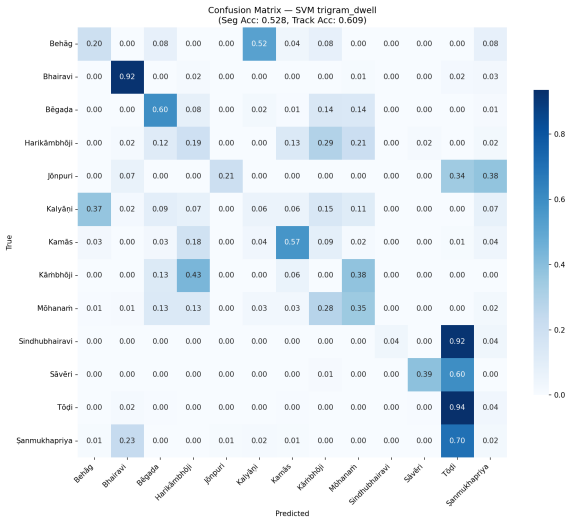
7 Stage 6: Classification

7.1 Primary Feature Sets

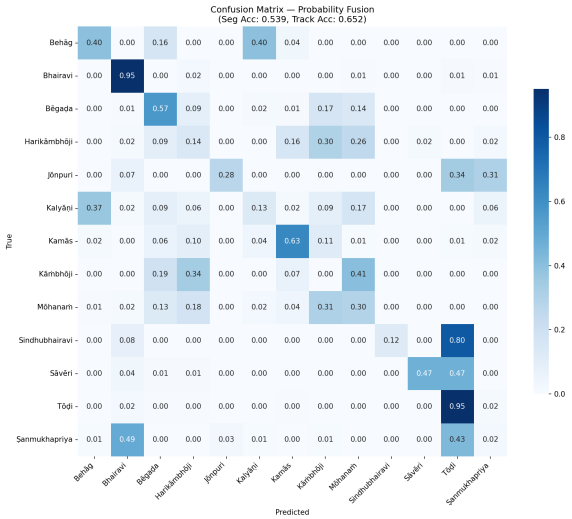


(a) Dwell + Vādi (16d) — 69.6% track acc, 67.8% F1w (b) Trigram (200d) — 63.0% track acc, 59.1% F1w

Figure 17: Confusion matrices for the two best-performing single-feature models. Dwell + vādi achieves the highest track-level F1w despite using only 16 dimensions.



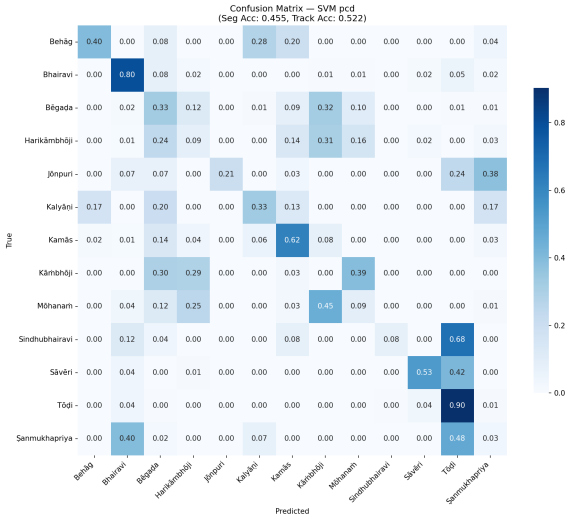
(a) Trigram + Dwell (216d) — 60.9% track acc, 57.2% F1w



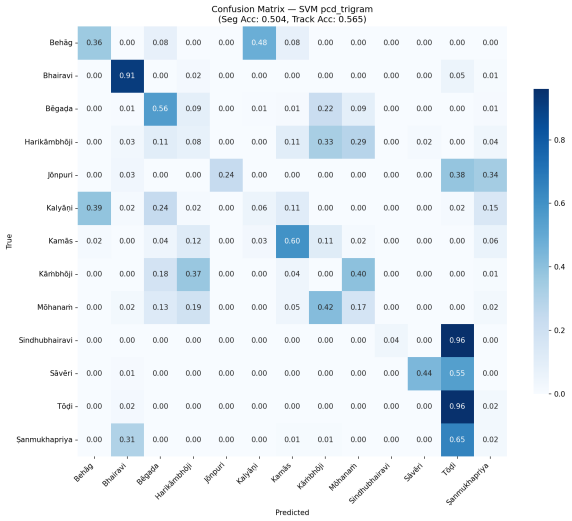
(b) Probability fusion — 65.2% track acc, 64.0% F1w

Figure 18: Combined feature models. Fusion improves over individual models but does not exceed dwell + vādi in track-level F1w, supporting cognitive parsimony (H3).

7.2 Additional Feature Variants

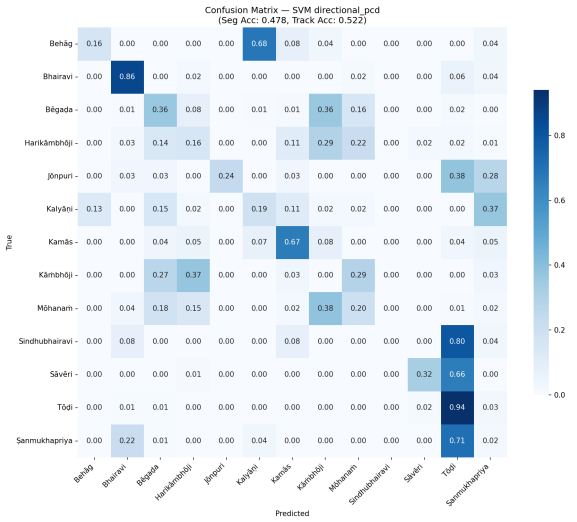


(a) PCD (120d) — 52.2% track acc, 49.6% F1w

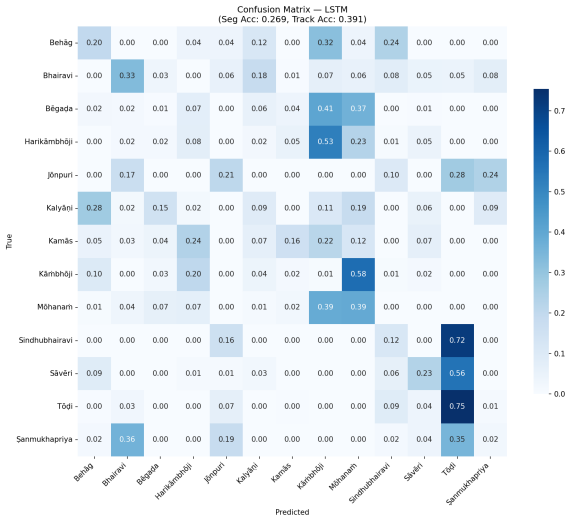


(b) PCD + Trigram (320d) — 56.5% track acc, 51.9% F1w

Figure 19: PCD-based confusion matrices. Adding PCD to trigrams does not improve over trigrams alone (H3: cognitive parsimony).



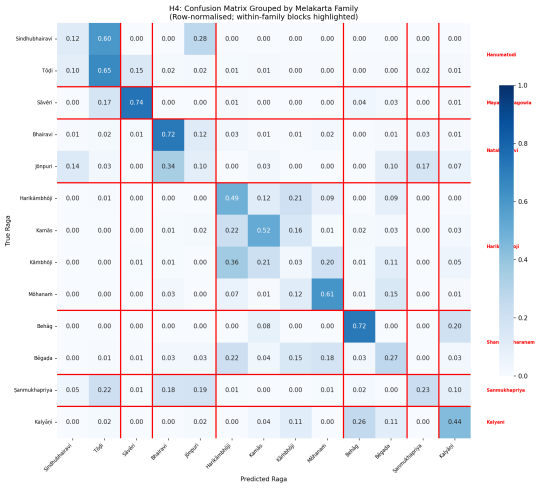
(a) Directional PCD (408d) — 52.2% track acc, 49.1% F1w



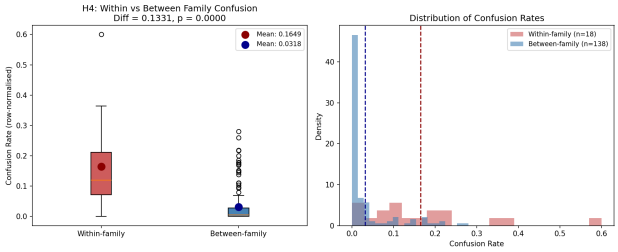
(b) BiLSTM — 39.1% track acc, 36.4% F1w

Figure 20: Directional features (H6, does not improve) and deep learning baseline. The BiLSTM suffers from data starvation (1,648 segments after filtering).

7.3 H4: Melakarta Family Confusion Analysis



(a) Confusion matrix annotated by melakarta family

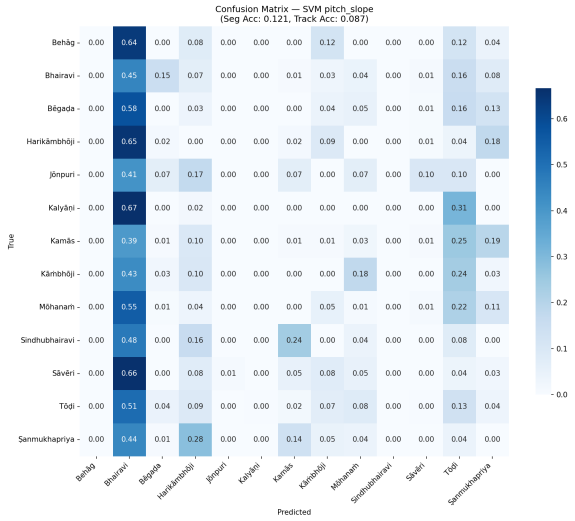


(b) Within vs between-family confusion ($p < 0.0001$)

Figure 21: H4 confirmed: within-family confusion is 5.19× higher than between-family under permutation test (10,000 iterations, $p < 0.0001$).

7.4 Negative Results

7.4.1 Gamaka Features

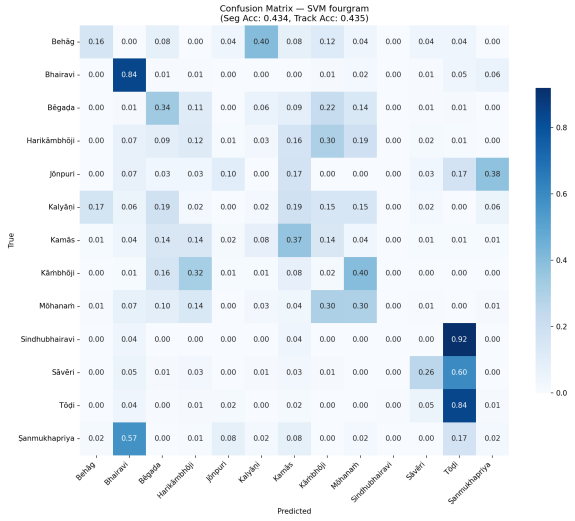


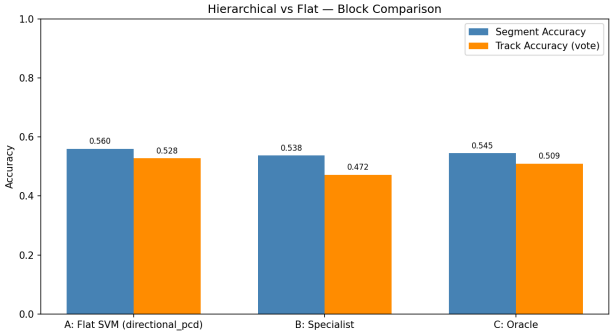
(a) Pitch slope features

(b) Combined gamaka features

Figure 22: Gamaka experiment. Continuous pitch contour features do not improve classification — pitch slope distributions are pan-rāga (all Carnatic music uses heavy ornamentation) rather than rāga-specific.

7.4.2 4-gram, Phrase Trigram, and Hierarchical





(a) 4-gram (200d) — 43.5% track acc, 38.8% F1w

(b) Hierarchical vs flat comparison

Figure 23: 4-grams (top-200 vocabulary): underperforms trigrams despite identical dimensionality, confirming longer n-grams are too sparse with ∼1,600 training segments. Hierarchical: flat baseline matches or exceeds hierarchy because 60.7% of errors are within-family (where L1 routing cannot help).

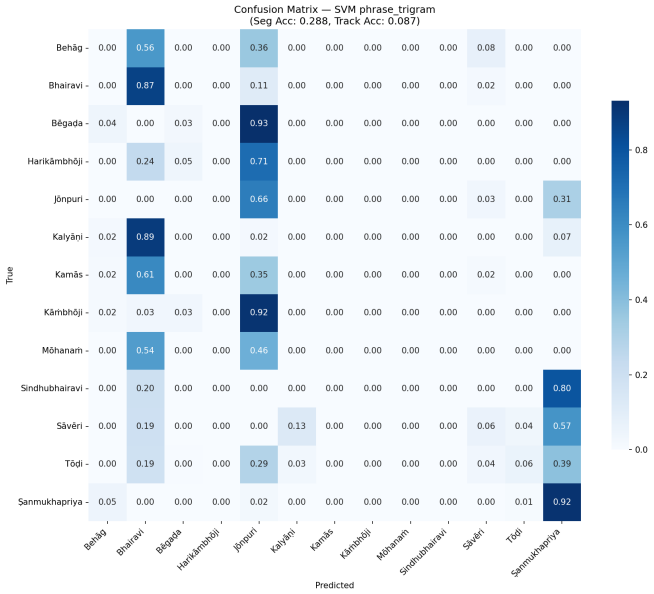


Figure 24: Phrase-aware trigram (200d) — 8.7% track acc, 2.8% F1w. Segmenting the token stream at silence/glide boundaries before computing trigrams collapses performance to below chance. The silence/glide boundaries that phrase segmentation discards are themselves informative for rāga identity.

7.5 Tonic Source Ablation: Full Estimated Tonic (All Tracks)

The table below shows classification results when **all tracks** are included using estimated tonic (no bad-tonic exclusion). This corresponds to `-run-mode est` in the pipeline. Comparison with the primary results (bad-tonic excluded) quantifies the downstream impact of tonic estimation errors.

Table 1: Classification results with estimated tonic, all tracks included (14 rāgas, 53 tracks, 2,107 segments). Compare with the primary results table in the main report (bad-tonic excluded) to see the downstream impact of tonic estimation failures.

Model	Seg Acc	Seg F1w	Trk Acc	Trk F1w
SVM trigram_dwell (216d)	43.38%	41.46%	56.60%	56.21%
SVM trigram (200d)	42.43%	40.72%	50.94%	50.86%
SVM directional_pcd (408d)	42.29%	39.83%	52.83%	49.33%
SVM directional_full (418d)	41.72%	39.32%	52.83%	47.91%
SVM pcd_trigram (320d)	42.15%	39.71%	47.17%	44.87%
SVM pcd_trigram_aux (330d)	41.29%	38.95%	49.06%	46.69%
SVM dwell_vadi (16d)	40.44%	37.83%	50.94%	45.21%
SVM pcd (120d)	37.07%	34.63%	50.94%	47.75%
SVM fourgram (200d)	36.97%	35.10%	43.40%	41.31%
Fusion (pcd+tri+tri_dwell)	44.19%	41.31%	56.60%	55.24%
LSTM swara_seq	36.73%	34.81%	43.40%	39.69%
SVM phrase_trigram (200d)	21.59%	17.93%	11.32%	5.04%
SVM mfcc (58d)	11.91%	3.95%	9.43%	3.45%

Key observations: (1) all models drop substantially compared to the bad-tonic-excluded condition, confirming that tonic estimation is the pipeline's primary bottleneck; (2) the ranking of feature sets is largely preserved — trigram_dwell and fusion remain top performers; (3) phrase_trigram collapses to 11.3% track accuracy (below random), confirming the negative result from the main report; (4) MFCC remains near chance (9.4%), consistent with H2 across conditions.

7.6 Tonic Source Ablation: Ground-Truth Tonic

The table below shows classification results when **ground-truth tonic** values from the SARAGA metadata are used instead of estimated tonic. This represents an upper bound on performance achievable if tonic estimation were perfect.

Table 2: Classification results with ground-truth tonic (13 rāgas, 53 tracks). Compare with the estimated-tonic results to isolate the impact of tonic estimation errors on downstream classification.

Model	Seg Acc	Seg F1w	Trk Acc	Trk F1w
SVM trigram_dwell (216d)	60.18%	59.50%	58.49%	54.90%
SVM trigram (200d)	57.95%	57.34%	56.60%	51.81%
SVM directional_pcd (408d)	56.00%	53.56%	52.83%	49.08%
SVM directional_full (418d)	55.10%	52.74%	50.94%	47.80%
SVM pcd_trigram (320d)	56.76%	55.14%	49.06%	45.71%
SVM pcd_trigram_aux (330d)	55.86%	54.38%	50.94%	47.01%
SVM dwell_vadi (16d)	52.97%	53.03%	52.83%	52.70%
SVM pcd (120d)	50.59%	49.47%	45.28%	40.25%
SVM fourgram (200d)	49.03%	48.91%	45.28%	42.13%
Fusion (pcd+tri+tri_dwell)	59.09%	57.82%	52.83%	49.31%
LSTM swara_seq	45.94%	45.15%	41.51%	39.44%
SVM phrase_trigram (200d)	17.89%	17.07%	9.43%	4.31%
SVM mfcc (58d)	11.91%	3.95%	9.43%	3.45%

Key observations: (1) segment-level accuracy improves substantially with ground-truth tonic (trigram_dwell: 60.2% vs 52.8% excl-est), confirming tonic quality is the primary bottleneck; (2) surprisingly, track-level accuracy for dwell_vadi is *lower* with ground truth (52.8%) than with estimated tonic excluding bad tracks (69.6%) — this is because the GT condition includes all 53 tracks (including difficult ones that were excluded in excl-est), while excl-est retains only 46 easier tracks; (3) MFCC remains at chance (9.4%), confirming H2 is robust across all tonic conditions.

8 Experiments

8.1 HPSS Tonic Estimation

Harmonic-Percussive Source Separation (HPSS) via librosa was tested as a pre-processing step for tonic estimation (`experiments/hpss_tonic_estimation.py`). The hypothesis was that isolating the harmonic component would improve pitch histogram clarity by removing percussive transients. Results showed no improvement over running CREPE on the original mixture — CREPE's neural pitch tracker is already robust to percussive interference.

8.2 Demucs Source Separation

Deep source separation using Meta's Demucs (HTDemucs model) was tested to extract clean vocals before pitch extraction (`experiments/demucs_tonic_estimation.py`). While Demucs successfully isolates vocals, the resulting tonic estimates did not improve over the mixture baseline — the computational cost ($\sim$10$\times$ slower) is not justified for this pipeline.

8.3 Hindustani Cross-Tradition Pilot

As a generalisability probe, the pipeline was applied to 10 tracks from the SARAGA Hindustani dataset (5 raags).

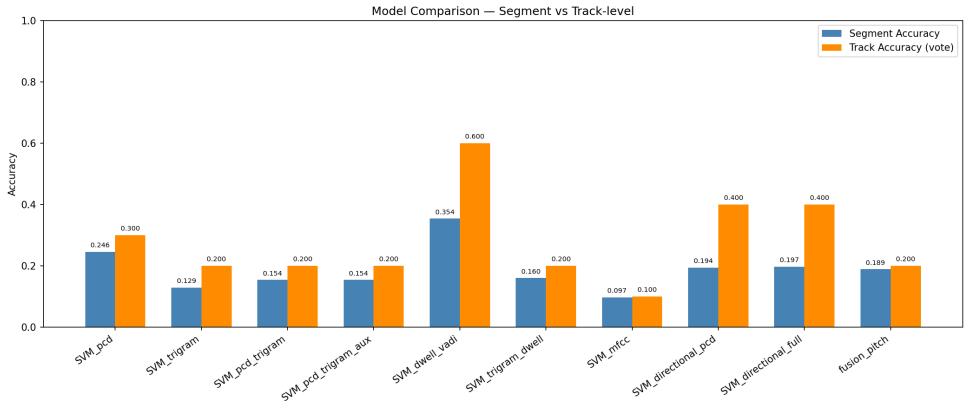
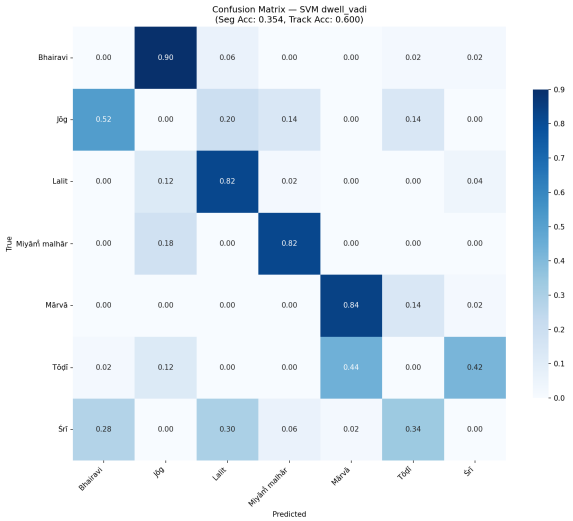
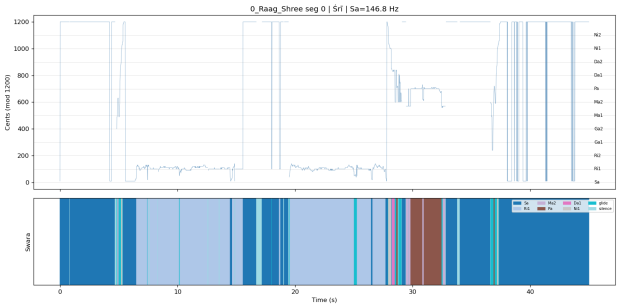


Figure 25: Hindustani model comparison (10 tracks). Feature-importance flip: dwell_vādi dominates while trigrams collapse — consistent with Hindustani music being emphasis-driven rather than phrase-driven.



(a) Hindustani dwell_vādi



(b) Hindustani swara mapping: Raag Shree

Figure 26: Hindustani results: dwell_vādi confusion matrix and swara mapping, demonstrating the pipeline operating on a different tradition.